

A BRIEF OVERVIEW OF AI

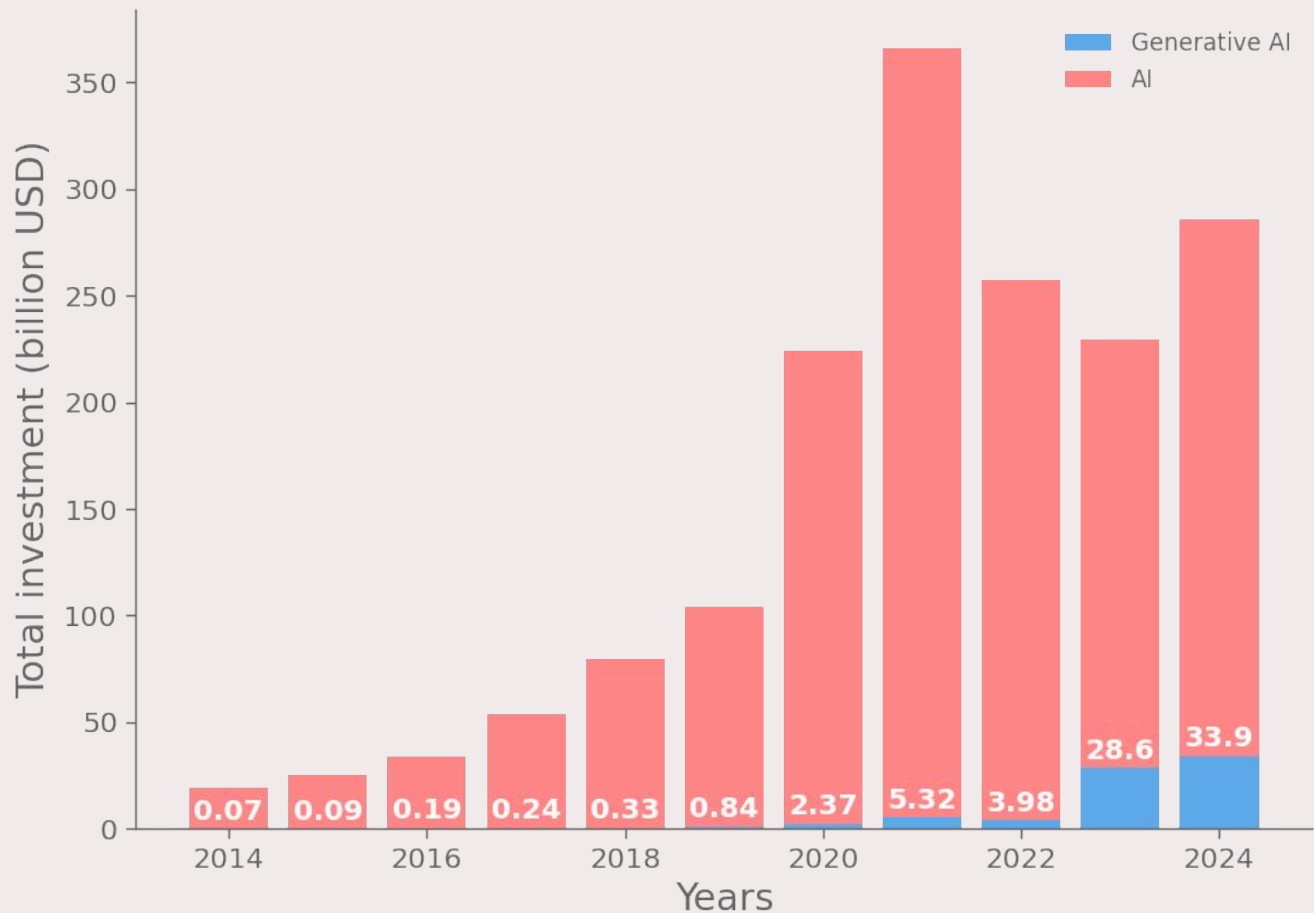
A sector by sector analysis

Giovanni Novati 68642A

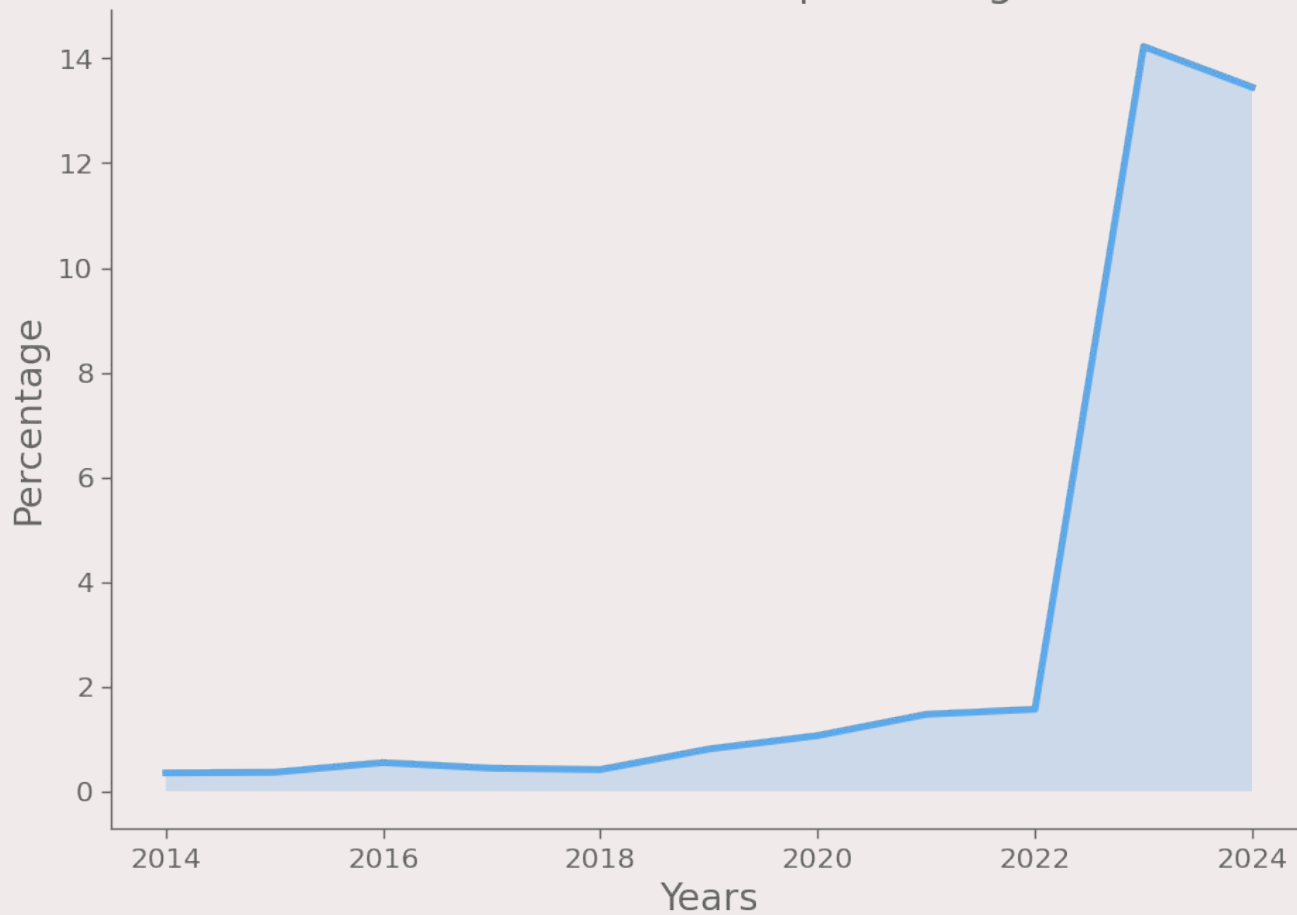
Simone Alessandro Casciaro 76678A

ECONOMY

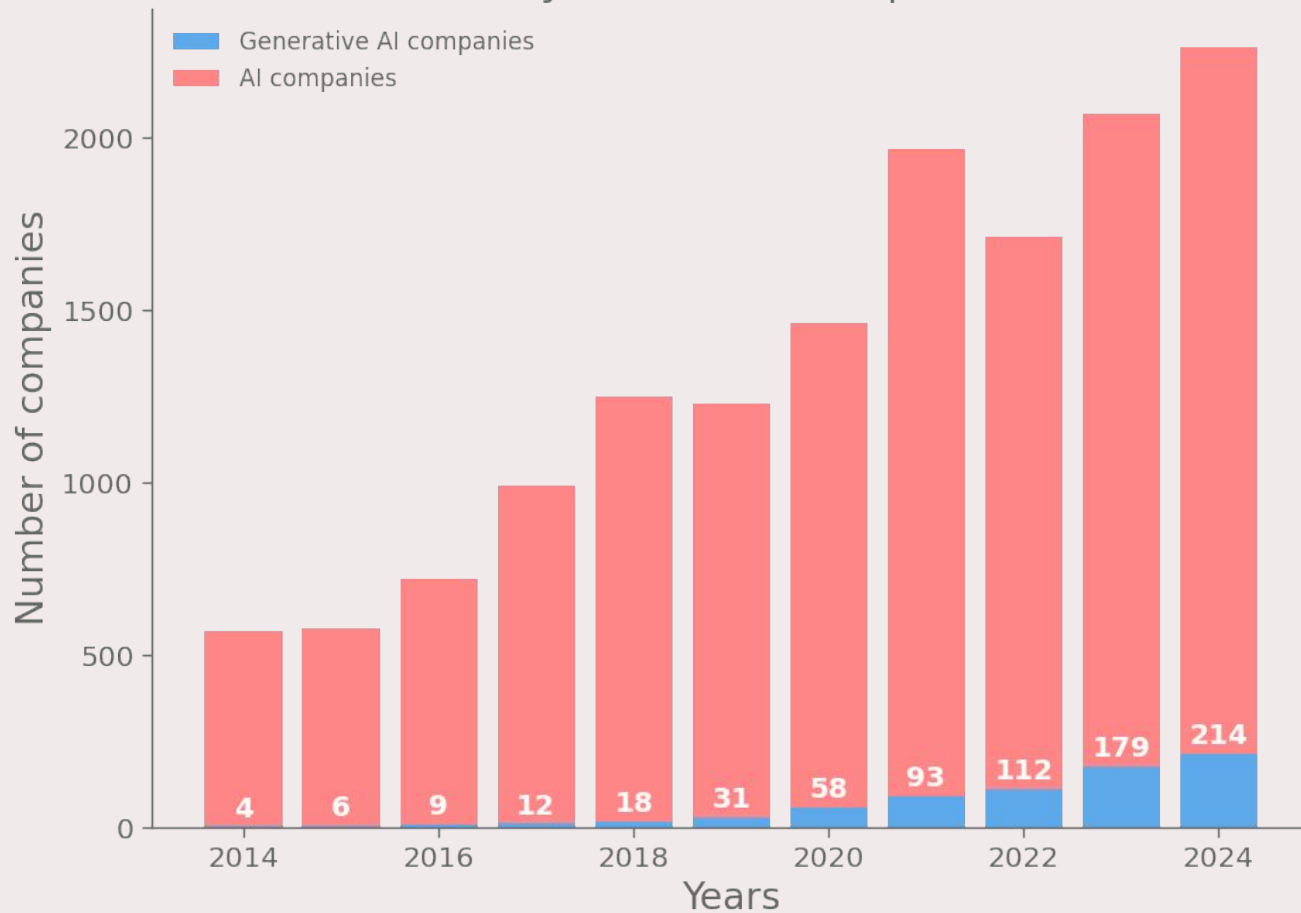
Global investment in the AI sector



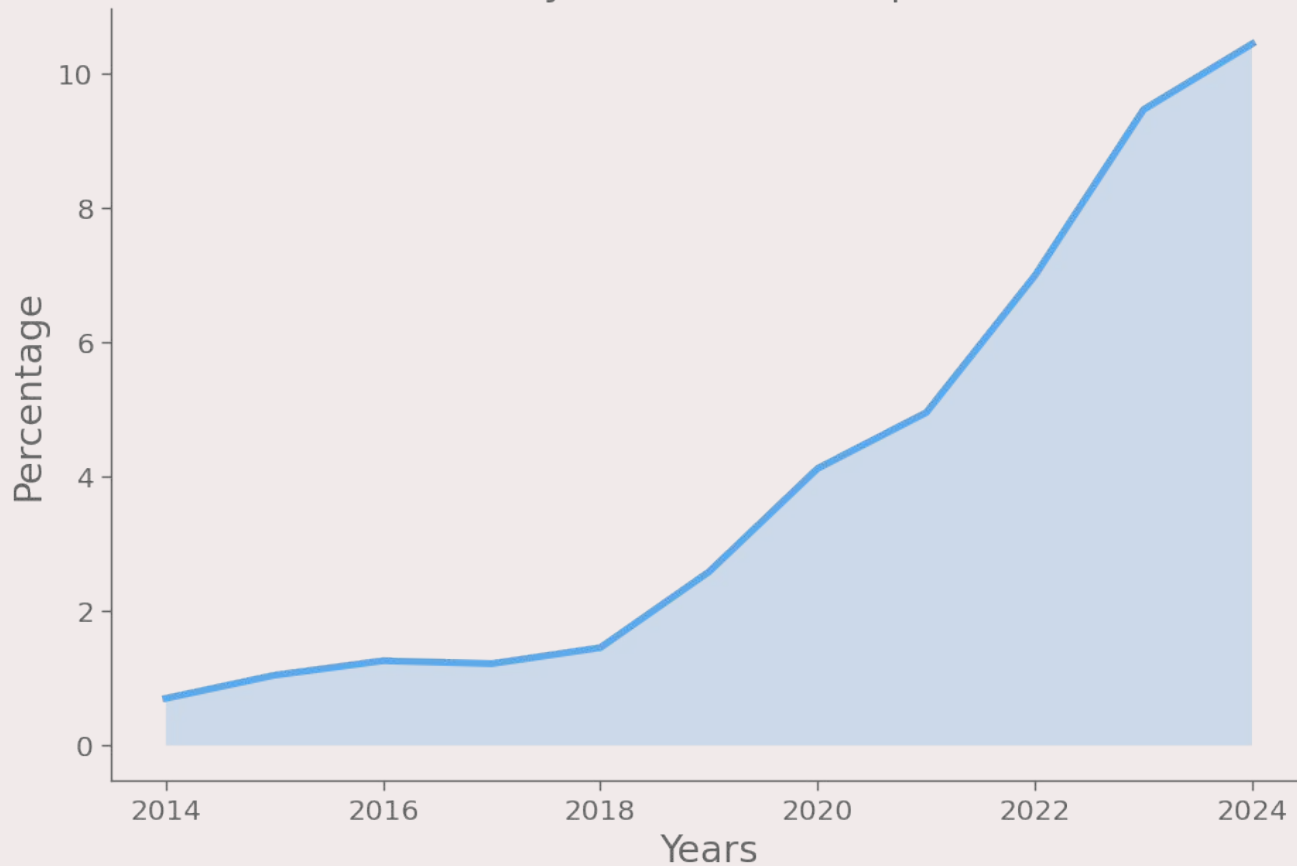
Generative AI investments as percentage of total AI



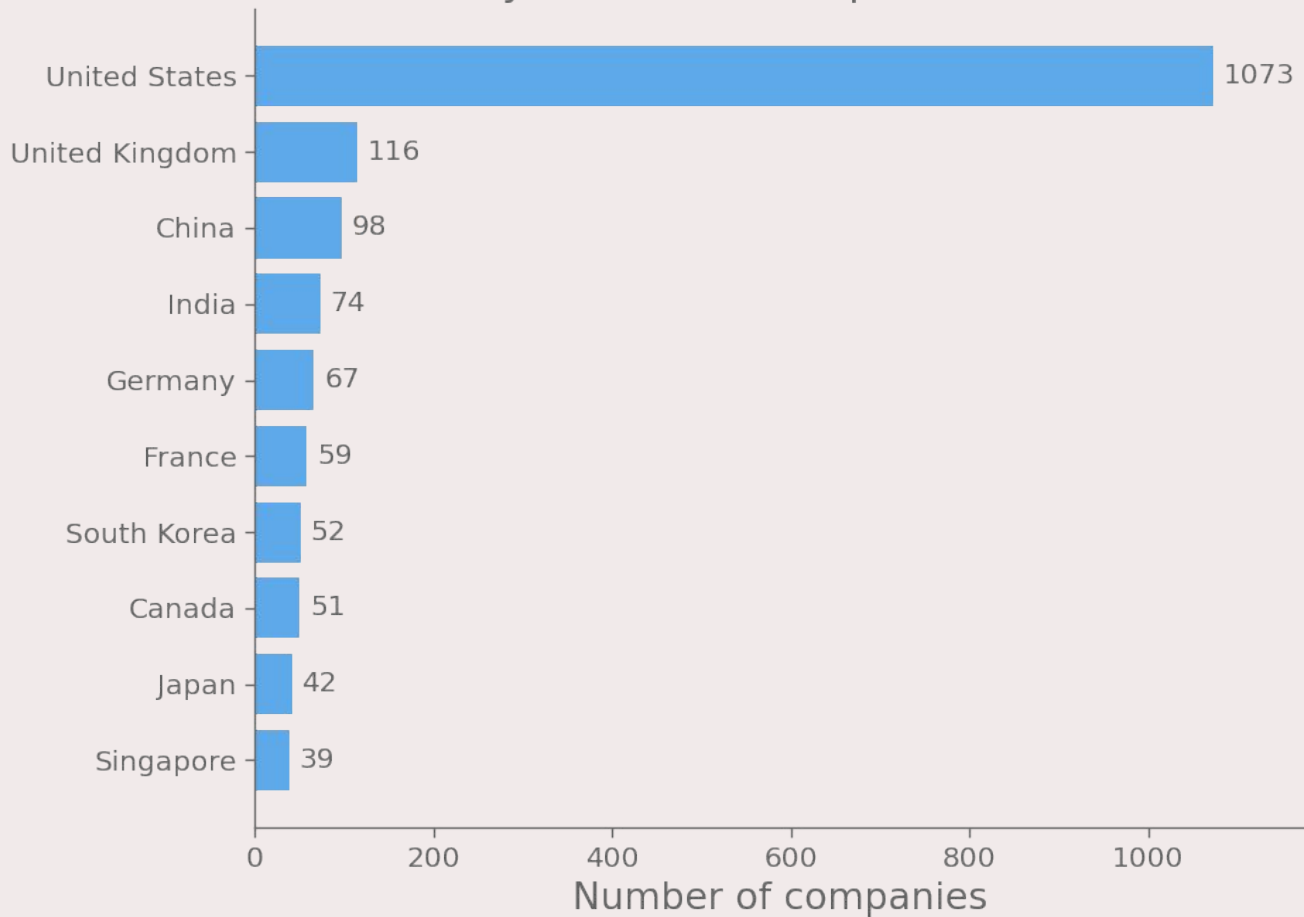
Newly funded AI companies



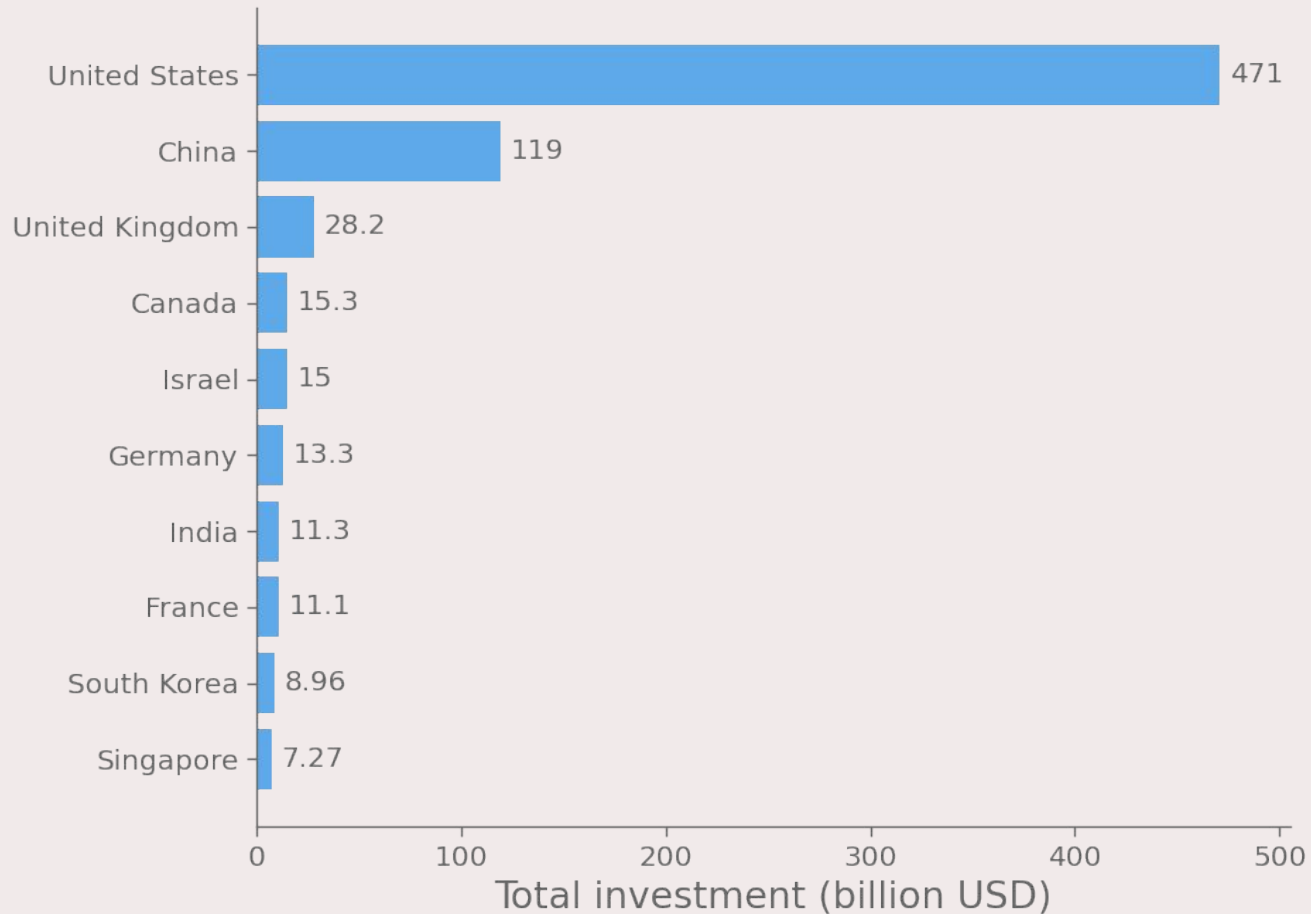
Generative AI companies as percentage
of newly funded AI companies



Newly funded AI companies in 2024

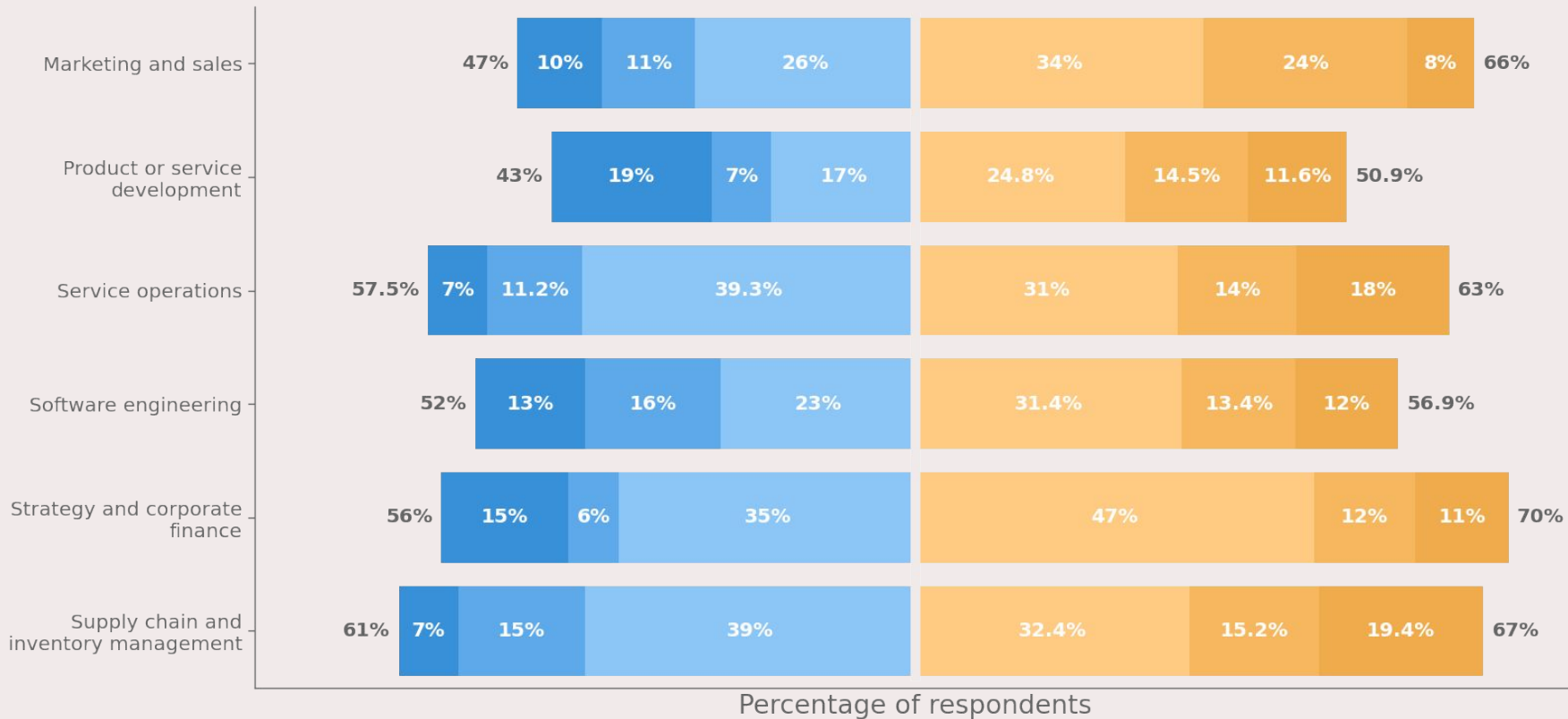


Private investments in AI from 2013 to 2024



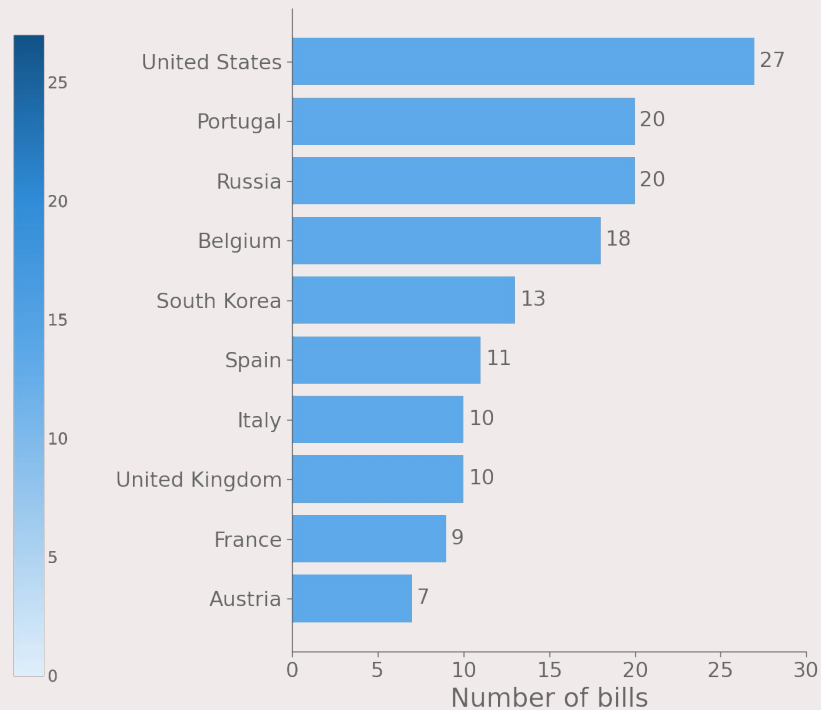
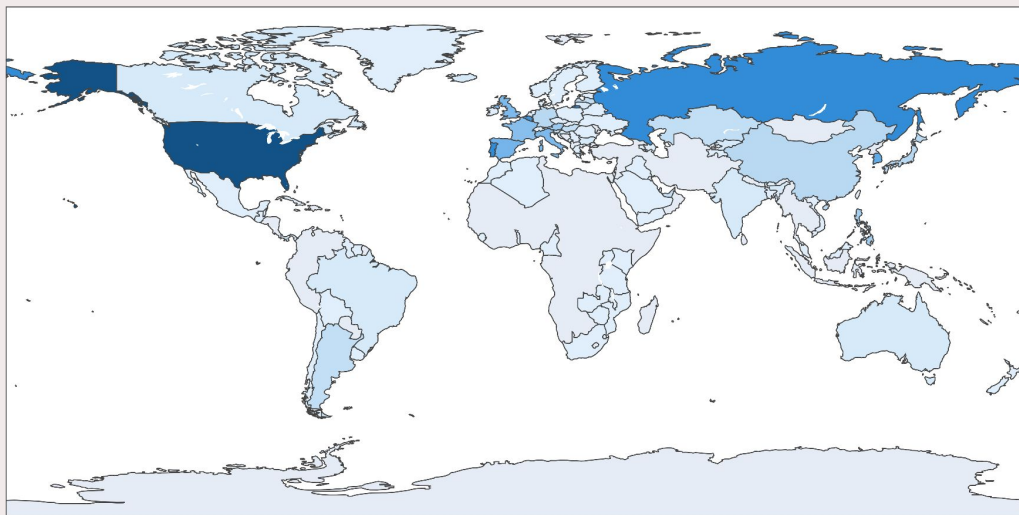
Company cost reductions and revenue gains thanks to generative AI in 2024

■ Decrease by ≥20% ■ Decrease by 11-19% ■ Decrease by ≤10% ■ Increase by ≤5% ■ Increase by 6-10% ■ Increase by >10%

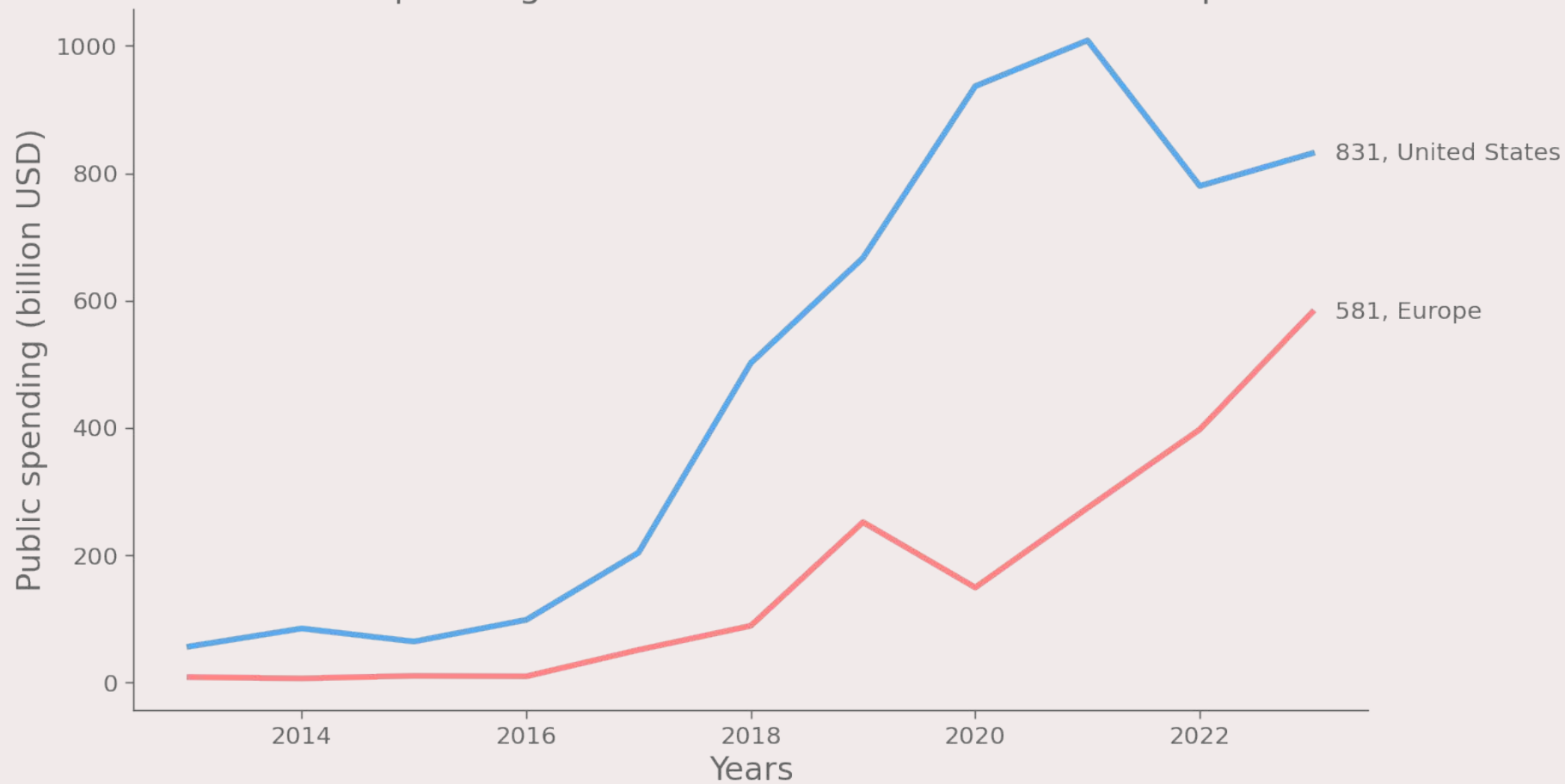


POLICY AND GOVERNANCE

Number of AI-related bills passed into law from 2016 to 2024



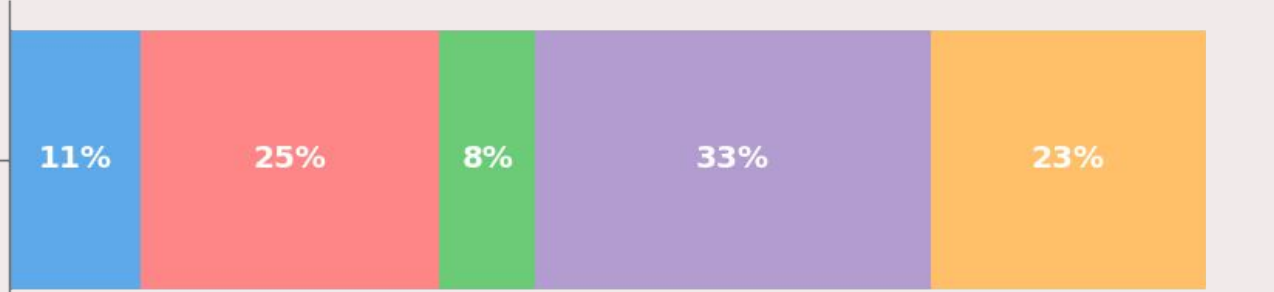
Public spending on AI-related contracts USA vs Europe



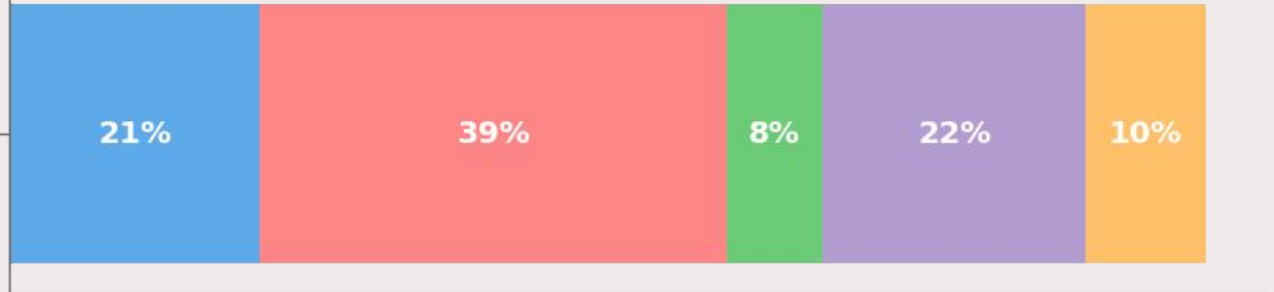
Global perceived opinion of AI on current jobs

Very likely Somewhat likely Don't Know Not very likely Not likely at all

AI will replace your current job in the next 5 years

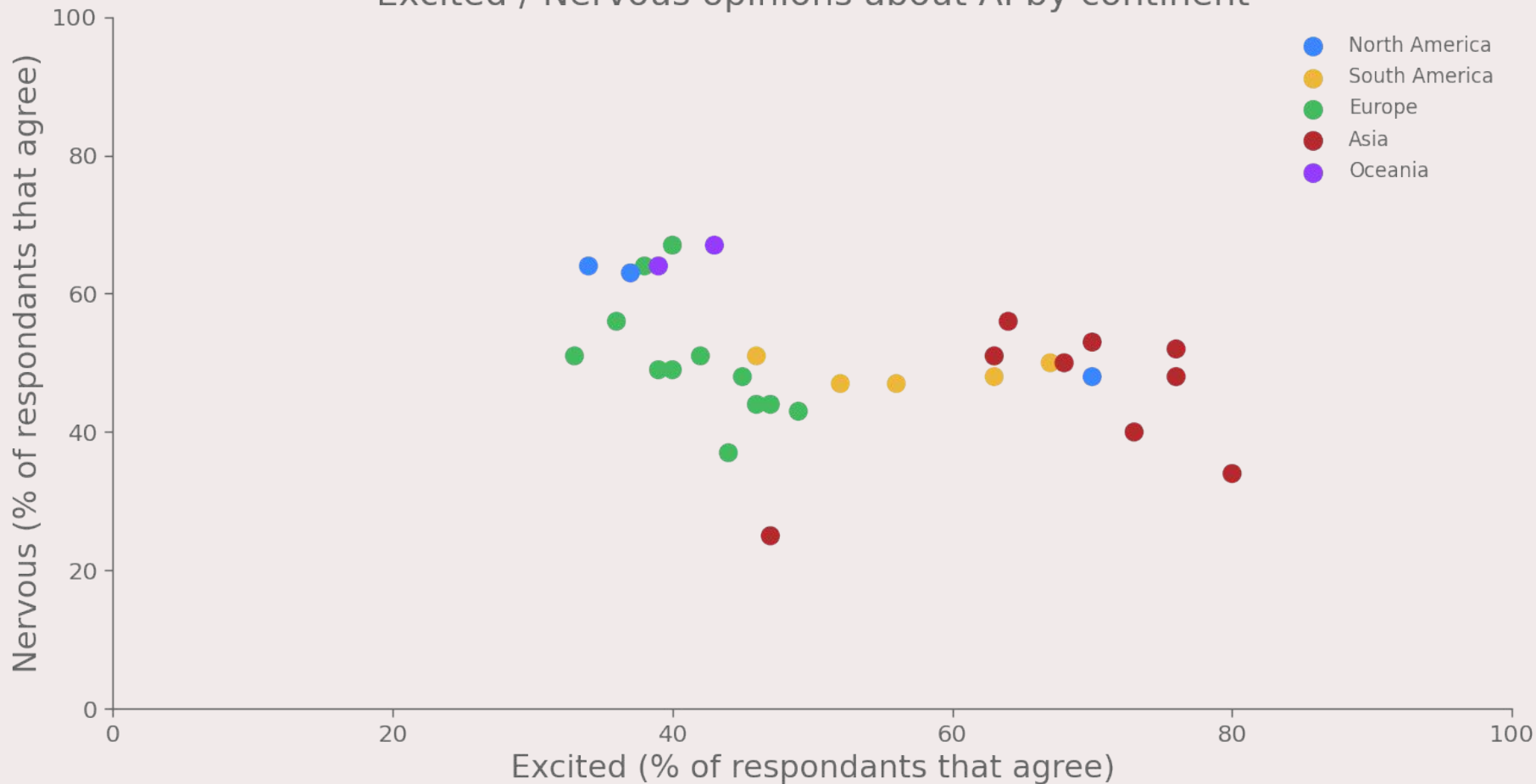


AI will change how you do your current job in the next 5 years



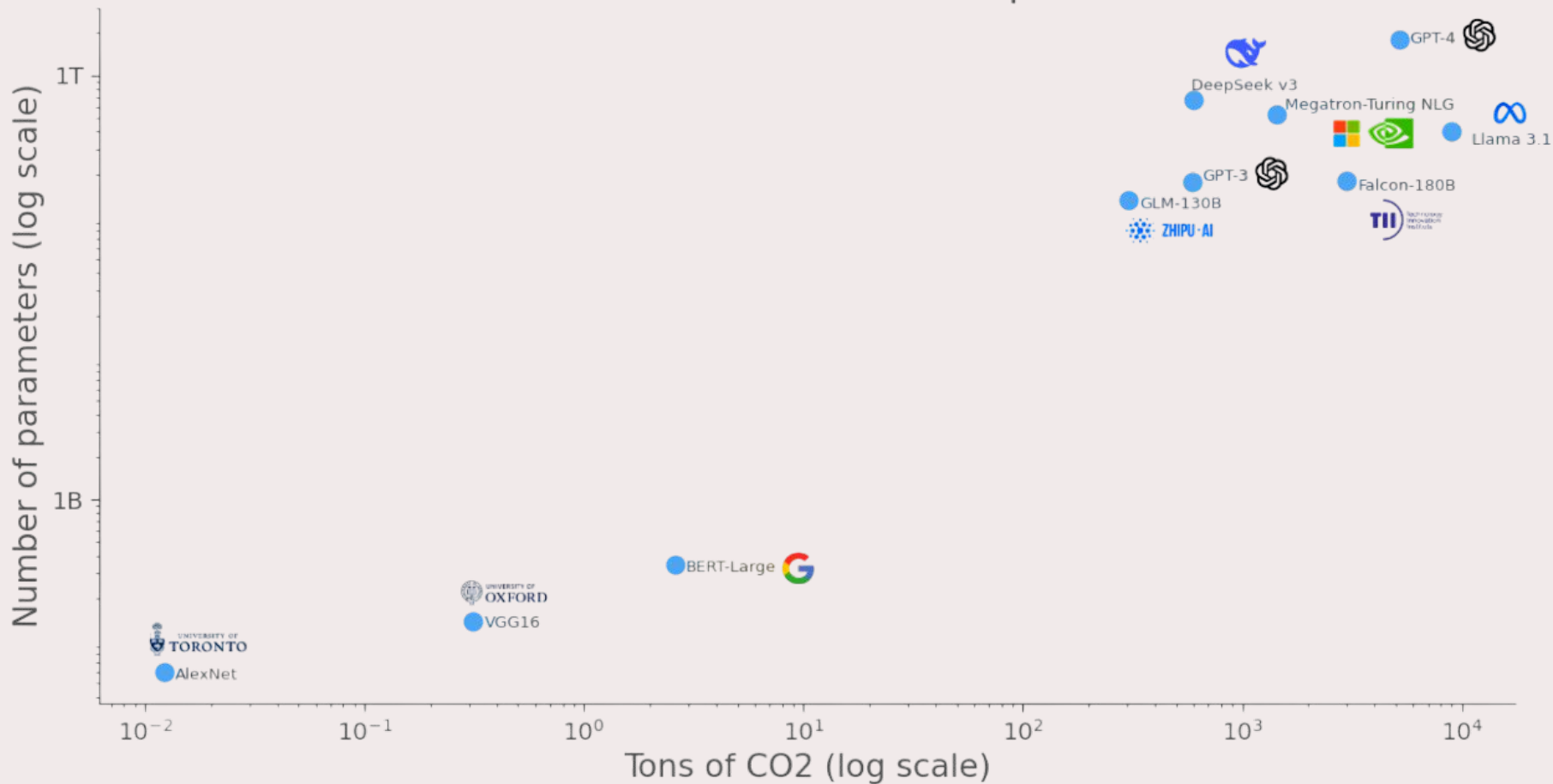
Percentage of respondents

Excited / Nervous opinions about AI by continent

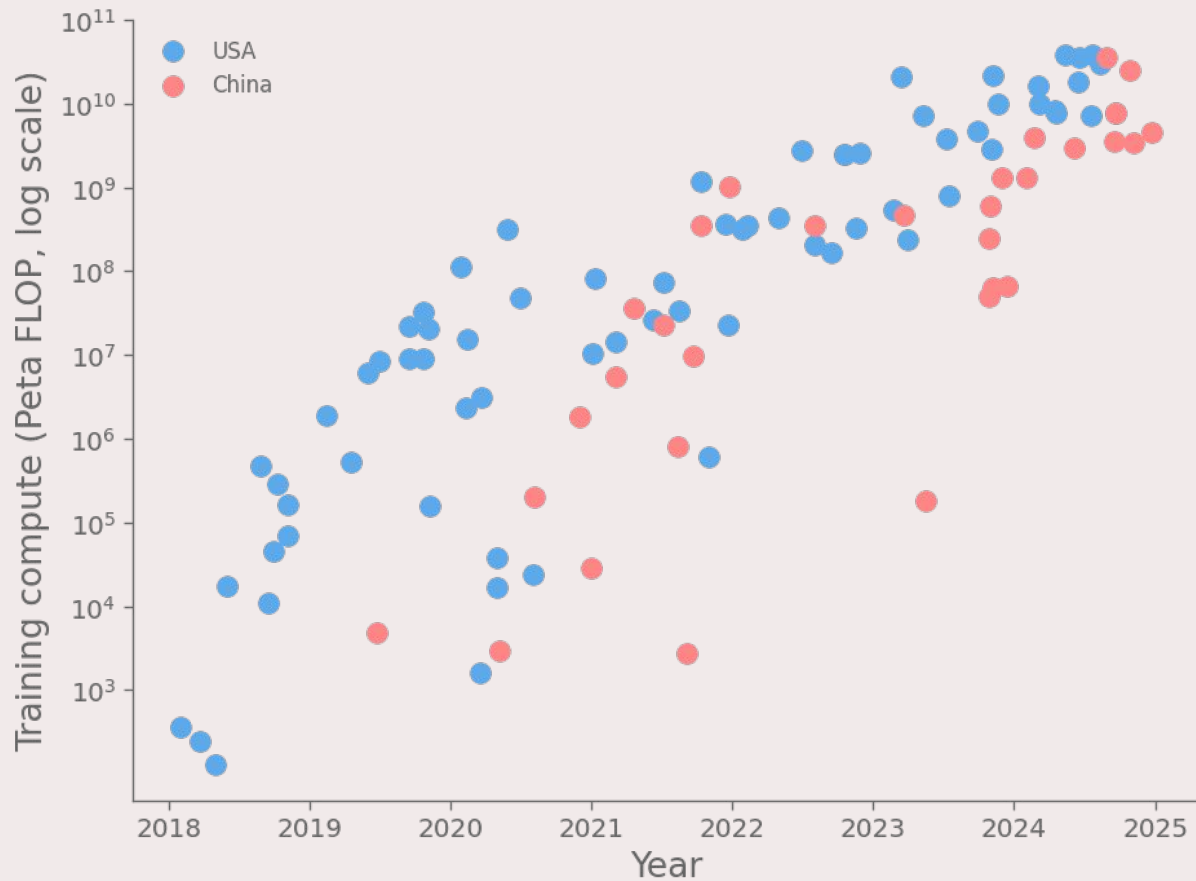


RESEARCH AND DEVELOPMENT

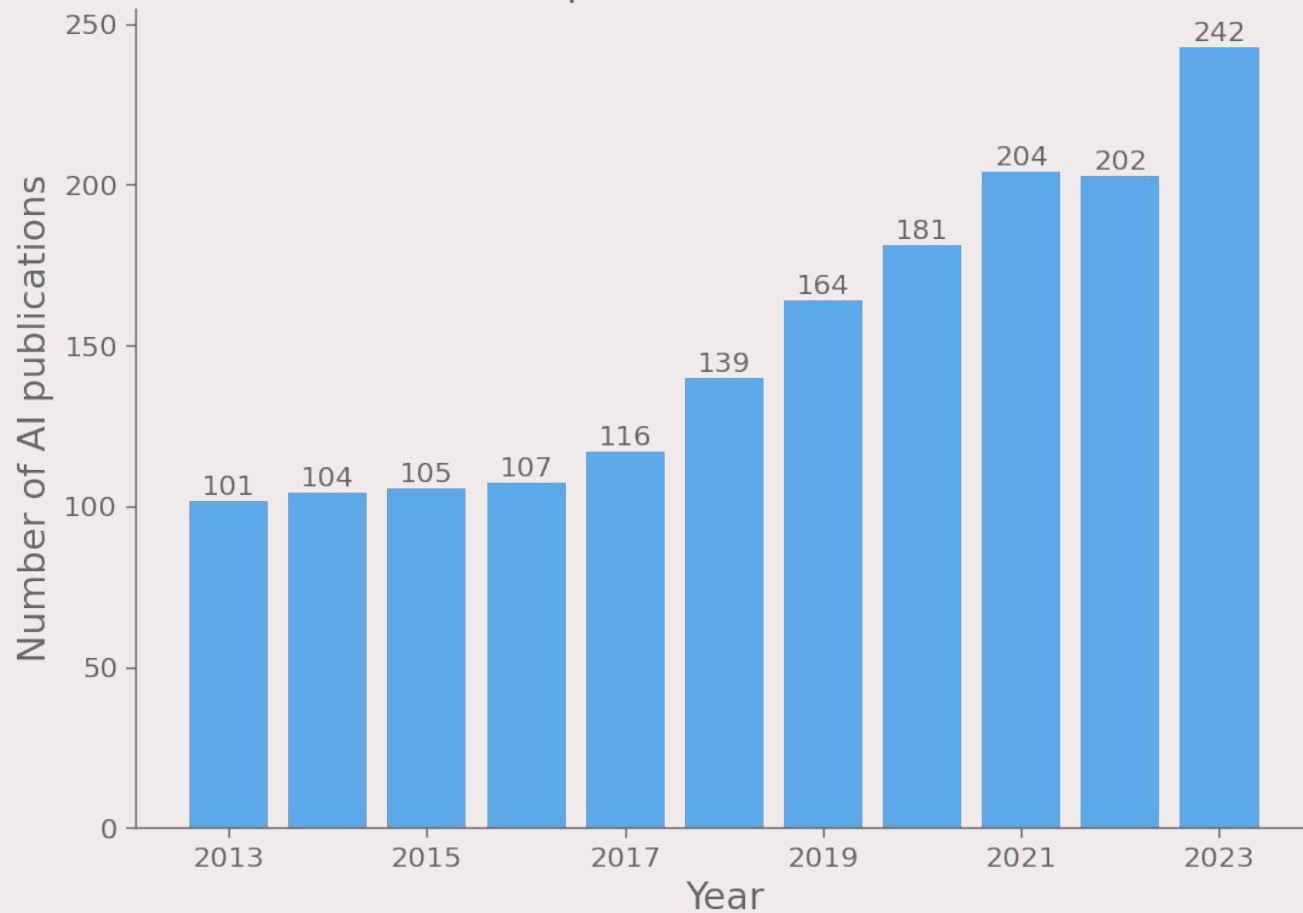
Carbon emissions and number of parameters



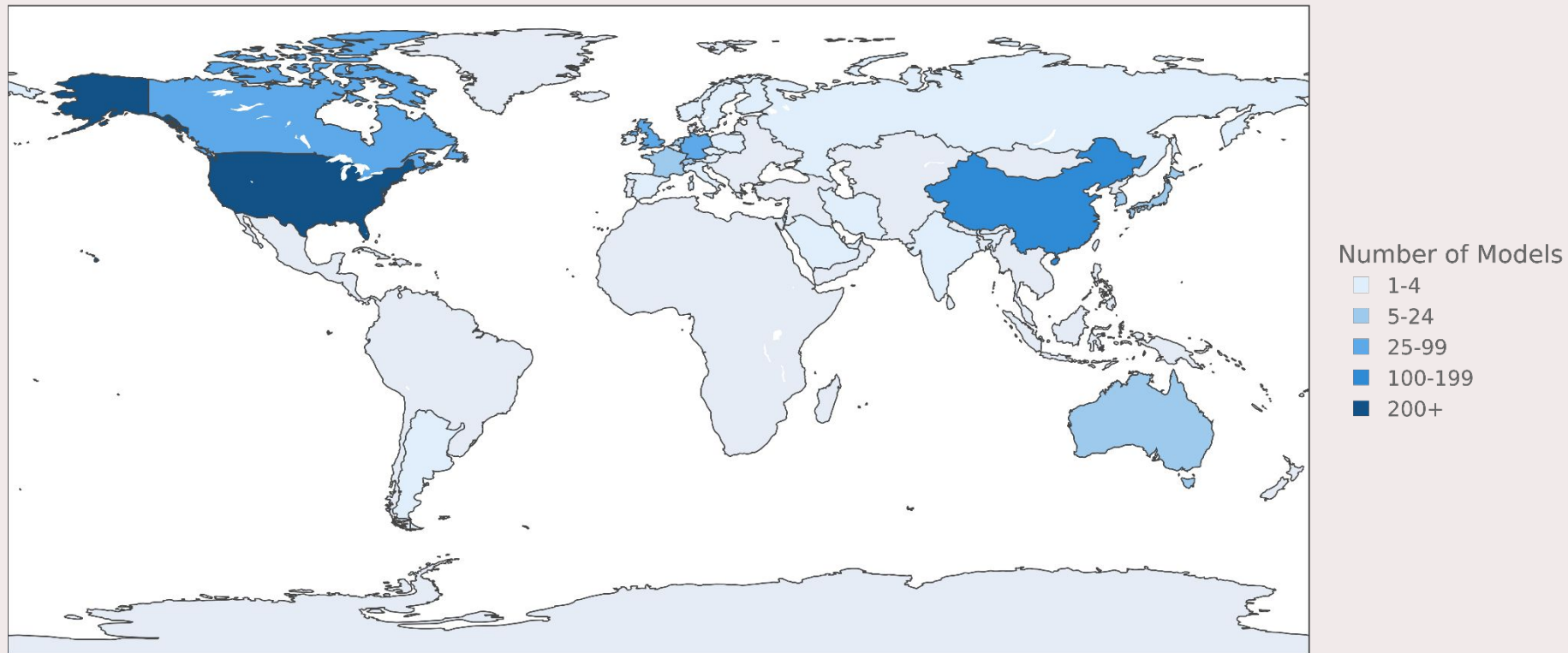
Training compute of select notable AI models in the United States and China



Number of AI publications in CS worldwide

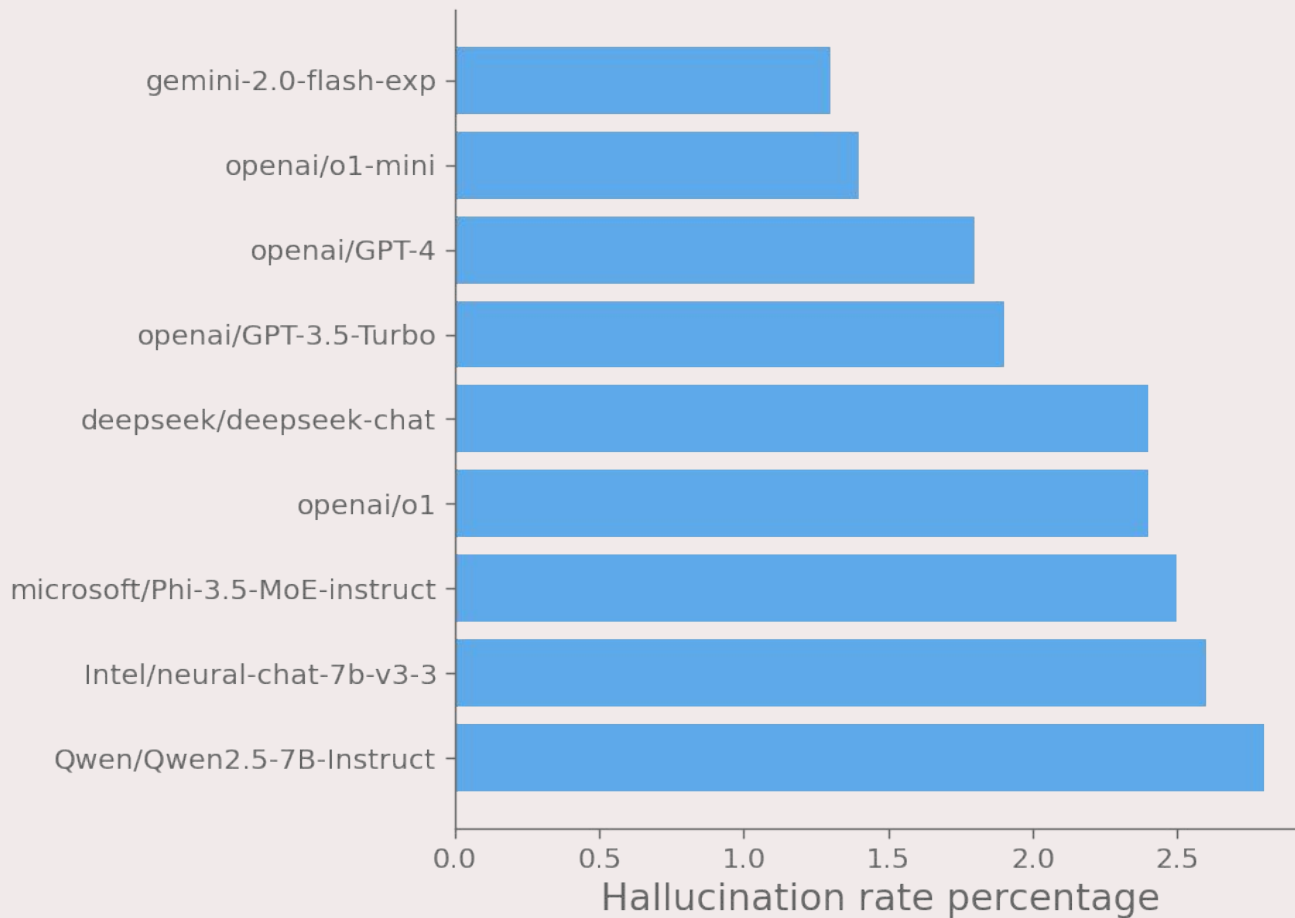


Number of Notable AI models from 2003 to 2024

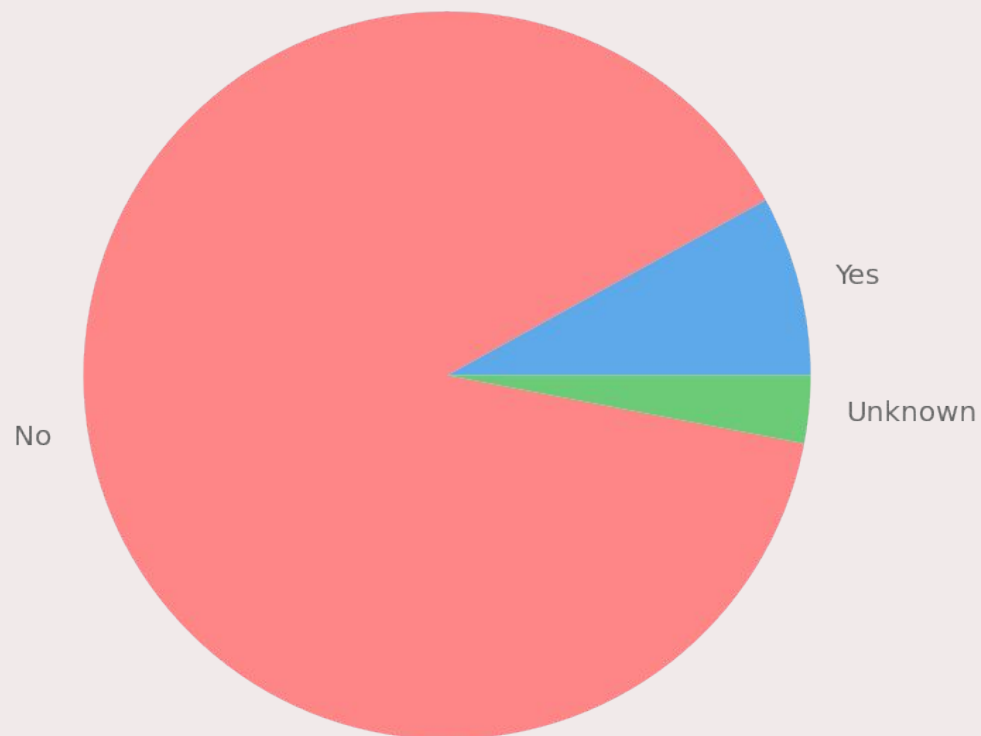


RESPONSIBLE AI

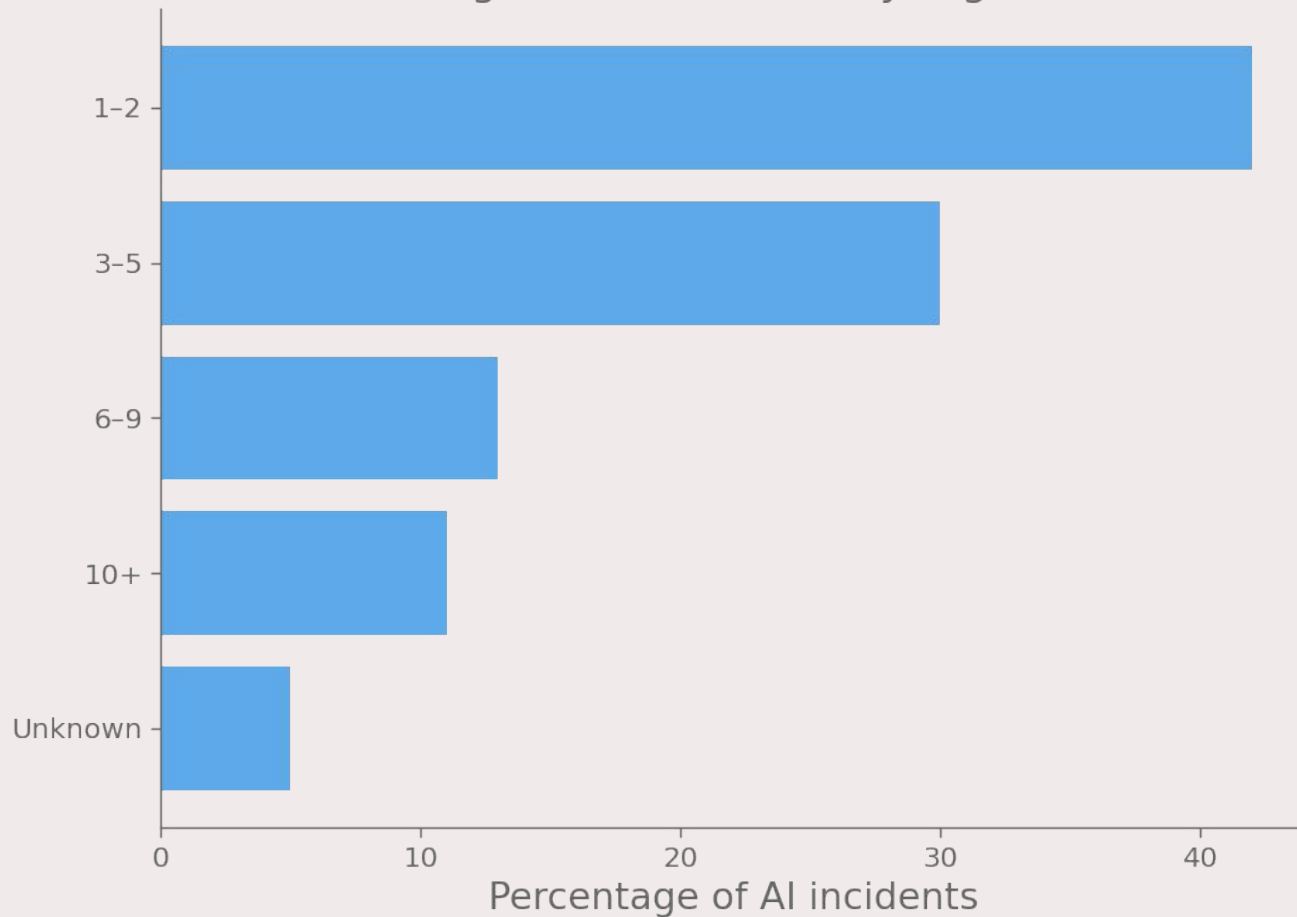
Hallucination rate of the latest models



Percentage of organizations who experienced AI incidents

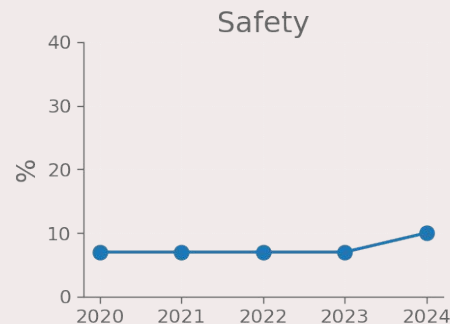
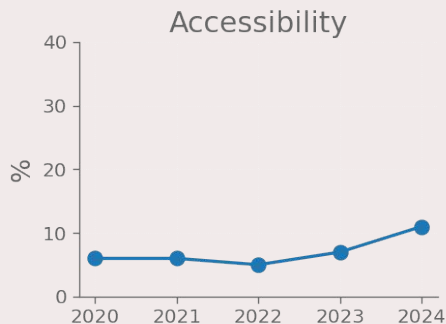
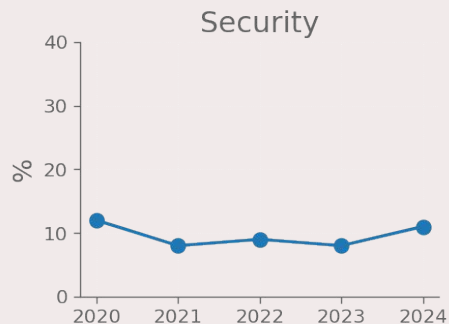
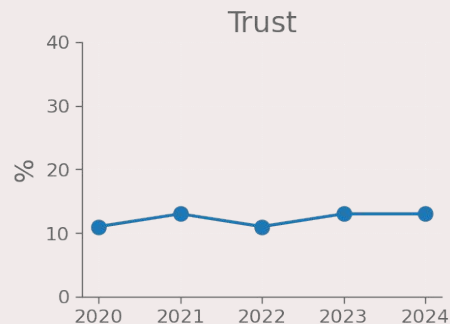
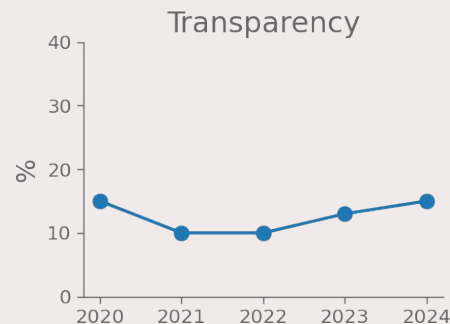
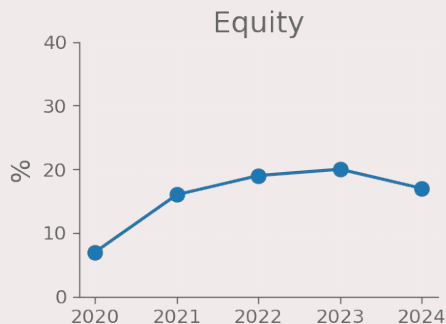
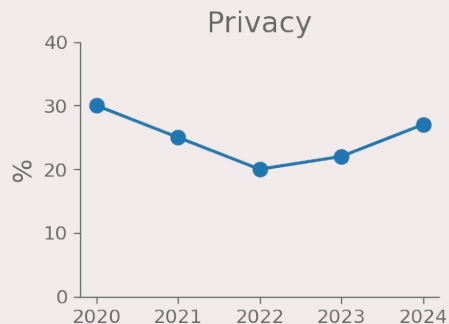
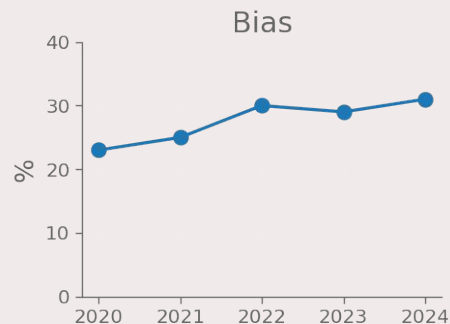


Percentage of AI incidents by organization

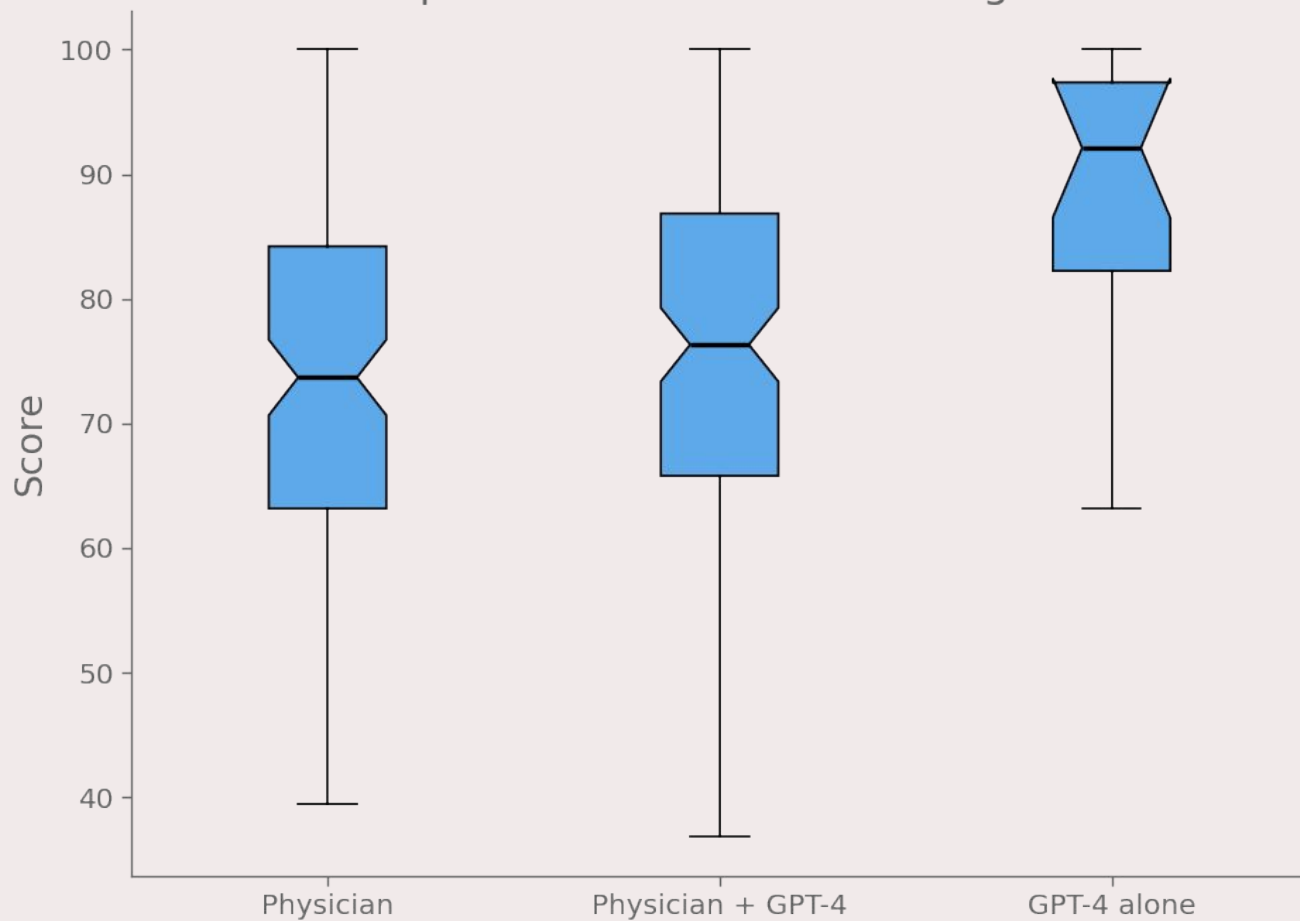


SCIENCE AND MEDICINE

Top 8 ethical concerns discussed in medical AI ethics publications

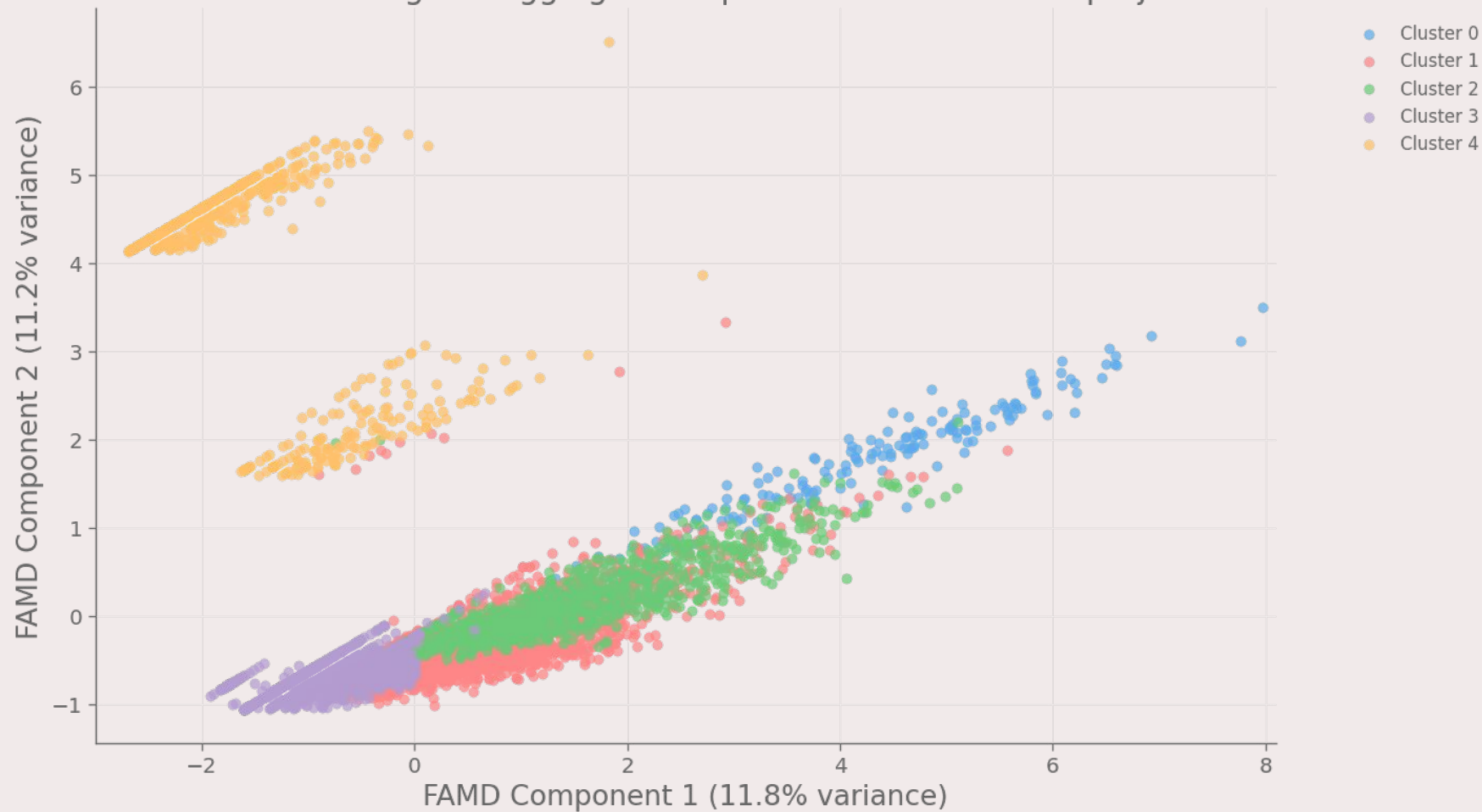


LLM performance in clinical diagnosis



CLUSTERING

K-Means clustering of HuggingFace top 7500 models - FAMD projection



Cluster	Size	Avg. Downloads	Avg. Likes	Top Architecture	Top Task
0 (●)	176	746.512	59	sentence-transformers	sentence-similarity
1 (●)	3.319	54.139	41	transformers	text generation
2 (●)	894	169.927	56	transformers	other
3 (●)	2.456	1.215	2	transformers	unknown
4 (●)	655	277.425	3	timm	image classification

Cluster	Downloads			Likes		
	Average	Median	SD	Average	Median	SD
0 (●)	746.512	4.522	7.316.683	59	11	318
1 (●)	54.139	1.608	1.032.386	41	7	147
2 (●)	169.927	2.042	1.511.338	56	10	191
3 (●)	1.215	665	1.822	2	0	4
4 (●)	277.425	2.266	4.781.000	3	0	7

Download distribution per cluster



Like distribution per cluster



Cluster	Wilcoxon's Test	
	Downloads	Likes
0 (●)	5.4e-20	2.5e-17
1 (●)	4.3e-52	1.7e-239
2 (●)	3.3e-37	2.6e-82
3 (●)	1.1e-261	0
4 (●)	4.68e-29	7.7e-52

Feature	Contributes to FAMD dimensions		
	Dimension 1	Dimension 2	Total
Architecture	0.319	0.464	0.784
Task	0.226	0.447	0.675
Likes	0.281	0.000	0.281
Downloads	0.173	0.088	0.261

Percentage distribution of tasks per cluster

Task	Clusters				
	CL 0	CL 1	CL 2	CL 3	CL 4
automatic speech recognition	0	6.2	4.8	1.4	0
feature extraction	15.3	1.5	0	0.2	0.3
text generation	1.1	68.8	4.1	0.2	0
image classification	0	0.3	0.2	0	96.9
token classification	2.3	8	0.6	0.4	0
sentence similarity	52.3	0	0	0	0
text classification	2.3	8	0.6	0.4	0
other	26.7	0	71.3	1.5	2.7
unknown	2.3	6.1	5.4	96.3	0

Percentage distribution of architectures per cluster

Architecture	Clusters				
	CL 0	CL 1	CL 2	CL 3	CL 4
peft	0	0	1	8.1	0
sentence transformers	87.5	0	0	0	0
timm	0	0	0	0	77.9
transformers	4	84.9	57.8	80.5	21.8
other	2.3	0	36	0	0
unknown	6.2	15.1	5.1	11.4	0.3

SOURCES

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THANKS FOR THE ATTENTION